

MECEE-BL (CEE) Syllabus - Medical Entrance

Medical Education Commission, Nepal - compiled from entrance.meded.gov.np (verified 2026-08-22)

Exam Overview

MECEE-BL (Common Entrance Exam — Bachelor Level), Medical Education Commission, Nepal

Mode: CBT | Duration: 3 hours | Questions: 200 MCQs | Total: 200 marks

Negative marking: 0.25 per wrong answer | 50th percentile = merit list eligibility

Registration: entrance.meded.gov.np | Conducted annually (2082 exam: 31 Oct - 4 Nov 2025)

Eligibility: 50%+ in +2 with Physics, Chemistry, Biology (min 2.4 GPA), age 17+ by Aug 31

Programs: MBBS, BDS, BSc Nursing, B.Pharm, BPT, BAMS, BNS, BSc MLT/MIT, BPH, BASLP, Optometry, Perfusion Tech

1. Physics (50 questions)

Mechanics: units, kinematics, dynamics, work-energy, circular & rotational motion, gravitation, fluid mechanics (pressure, buoyancy)

Heat & Thermodynamics: thermal expansion, calorimetry, gas laws, kinetic theory, laws of thermodynamics, humidity

Waves, Sound & Optics: wave motion, sound waves & Doppler, reflection, refraction, spherical mirrors & lenses, prism, dispersion, interference, diffraction, polarization

Electricity & Magnetism: charges, electric field & potential, capacitance, Ohm's law & circuits, magnetic fields, Oersted & Faraday experiments, induction, transformer basics

Modern Physics: electron, photoelectric effect, X-rays, radioactivity, nuclear reactions (fission/fusion), semiconductors

2. Chemistry (50 questions)

Physical Chemistry: atomic structure, periodic classification, chemical bonding, oxidation & reduction, electrolysis, chemical & ionic equilibrium, acids/bases/salts, mole concept & stoichiometry, solutions, thermodynamics, kinetics, colloids

Inorganic Chemistry: hydrogen, alkali & alkaline earth metals, halogens, noble gases, transition elements, coordination compounds basics, metallurgy, environmental chemistry

Organic Chemistry: functional groups, nomenclature, isomerism, hydrocarbons, alcohols & phenols, aldehydes/ketones, carboxylic acids, amines, aromatic compounds, polymers, biomolecules

3. Biology (100 questions)

Botany (50):

- Cell biology: structure, organelles, cell division (mitosis/meiosis), plant tissues
- Plant physiology: photosynthesis, respiration, transpiration, mineral nutrition, plant hormones
- Reproduction: vegetative, sexual in flowering plants, seed & fruit
- Genetics & evolution: Mendelian inheritance, DNA/RNA, mutation, Darwinism basics
- Diversity: algae, fungi, bryophytes, pteridophytes, gymnosperms, angiosperms; morphology & taxonomy

Zoology (50):

- Animal tissues, organ systems (digestive, respiratory, circulatory, excretory, nervous, endocrine)
- Human physiology: blood & lymph, heart, kidneys, lungs, neuron & synapse, hormones
- Genetics & evolution: human chromosomes, sex determination, genetic disorders
- Diversity: Protozoa to Chordata; classification, mammals & birds; animal behavior basics
- Health & disease: pathogens, immunity, vaccines, common diseases of Nepal

4. Mental Ability Test (MAT) — integrated

Logical reasoning, number series, analogy, classification, coding-decoding, spatial reasoning, basic math aptitude

Admission Flow

MECEE rank → open-house matching: candidate ranks college+program preferences, system matches by rank

Seats: 2,635 MBBS total (691 scholarship) for 2082 intake

Government colleges (IOM, BPKIHS, PAHS): heavily subsidized ~NPR 15-20 lakh total

Private MBBS: ~NPR 40-50 lakh total

Top ~100 ranks: government MBBS seats; top ~500: private MBBS/BDS; 1000-2000: allied health

Key dates 2083: exam ~Ashoj 2083 (early October 2026), classes from Magh 2083